# Same endpoints, different carbon origins

EnzymeMap 13175 · complete composition $C_4H_6O_5$ · all 9 heavy atoms shown

Formaldehyde · Carboxyl group · Remaining carbon chain · Break / weaken · Form / strengthen

## A Recovered reference mapping

**4 heavy-atom changes**

**Carbon 1 joins the three-carbon molecule; carbon 4 becomes $CO_2$.**

1 broken + 1 formed + 1 weakened + 1 strengthened

## B MAPPA alternative mapping

**2 heavy-atom changes**

**Carbon 1 becomes $CO_2$; carbon 4 stays in the three-carbon molecule.**

1 broken + 1 formed

Colors track atom origins; carbon numbers keep the same identity across both endpoints.

H atoms are implicit. With their assignments optimized, either heavy-atom mapping has 6 total events.